

Premo: Powerful Character Rigging, Fast Animation

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CCS CONCEPTS

• Computing methodologies → Animation.

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1 INTRODUCTION

Premo is an Academy Award winning animation and rigging application developed by DreamWorks. Premo's animation experience was presented at SIGGRAPH 2014 [Gong et al. 2014]. Premo's evaluation engine, LibEE, was presented at DigiPro 2012 [Watt et al. 2012] and LibEE version 2 at DigiPro 2018 [Bryson and Papp 2018] where a dual representation was introduced in order to enable the authoring requirements of rigging in Premo. This talk presents how we approached rigging within Premo, including a brand new data model and support for visual programming. While some tools in the industry have taken steps in this direction, many still depend on build scripts before a rig can be animated. Premo empowers artists to focus on the creative aspects of rigging as opposed to technical workflows. For example, with the introduction of Pose Centric Workflows, artists are able to rig directly using models in pose space. Premo runs editable, full-resolution production rigs while maintaining the interactive animation performance Premo is known for.

2 RIGGING IN PREMO

Premo's prior data model had no support for rigging workflows. Its focus was curve editing and animation, with basic support for other features such as constraints and drawings. While the data model

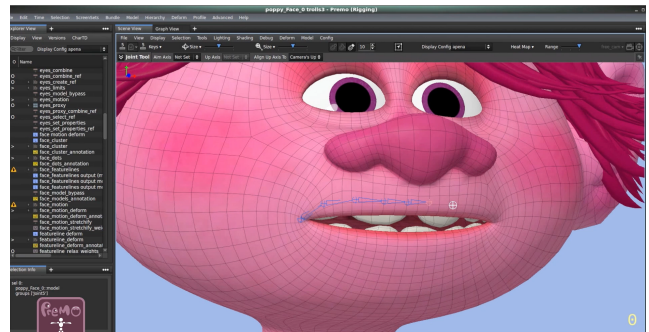


Figure 1: Property of DreamWorks

could support loading character rigs, it was not well suited for authoring a rig. The graph language was geared entirely towards fast evaluation and lacked various rigging features such as referencing, instantiation, derivation, and overrides. Premo's new data model provides all of this and more in support of a compelling rigging experience.

2.1 Premo Data Model

2.1.1 Definitions. At the core of the Premo data model is the ability to create modular definitions. A definition can be of any type, but the most common definition is a Bundle. Bundles are discrete containers of data, defined by the application's plugins, containing authored scene data such as graph nodes, models, curves and more. Other definition types include graph attribute types and drawings.

2.1.2 Versioning. Each time a definition is modified and saved, a new version is created. Unlike most other platforms where versioning is either non-existent or left to an external revision control system, Premo has native and robust integration of version management. From within Premo, the user can perform version related operations such as browsing the version history, inspecting meta-data such as author, comments, and publish labels, loading a specific version, or merging different versions together.

2.1.3 Reuse through Versioned Referencing. Definitions can reference other definitions through derivation and instantiation. This enables reuse, such as Hiccup deriving from a biped definition which instantiates a single arm definition twice, and in turn, each arm instantiates a finger definition four times. Importantly, a reference to another definition is to a specific version of that definition.

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This enables powerful workflows such as locking off portions of a character or shot from receiving updates, rolling back a reference to a previous version when a newer version introduces issues, or automatically updating the references to newer versions, for example when working on a branch with a colleague during character development.

2.1.4 Sparse Overrides. Referencing is useful enough on its own, but its real power comes from being able to sparsely modify subsets of the data contained within the referenced definition. As a simple example, a biped might instantiate the same arm definition twice, but one instance would need to be overridden to flip the axis of rotation of the elbow in order for the left and right arms to behave correctly relative to the side of the character. If a new version of the arm definition is published, the biped can update the left and right arm instances to point to this new version, automatically receiving all the benefits of this new version, while still maintaining the sparse overrides that were made to the elbow rotation. Premo's data model supports adding, updating, and deleting objects, as well as modifying properties of those objects. This includes the ability to override the definition an instance is pointing to, for example, replacing Hiccup's left leg with a peg leg.

2.1.5 Conflicts and Resolutions. Workflows involving sparse overrides can result in data inconsistencies. For example, overriding a value of an attribute in a rig, while a separate override removes the attribute in question. Premo's data model represents inconsistencies like these as *conflicts*, and enables users to resolve these conflicts in-app through graphical user interfaces and APIs. Plugin authors can implement custom conflict resolution strategies to aid in this process.

2.2 Bundle Workflows

Premo builds on top of its powerful data model by providing easy to use workflows to author characters. The Resource View provides artists with access to the entire catalog of definitions. Using this view, artists can now drag and drop definitions to assemble existing character modules to construct a new character in a fraction of the time. Common actions are easy to perform and do not require manually creating nodes and connections in the Dependency Graph View, while more advanced workflows are still possible. Additionally, animators can now make in-shot rigging 'on-demand' changes, such as adding deformers on a shot-specific basis.

2.3 Pose Centric Workflows

Premo provides pose-centric workflows that enable character artists to directly sculpt the output of deformers in any pose. Premo maps those edits back to the underlying pose shapes, taking weighting into account and ensuring that the edits can blend seamlessly into the pose range. This enables artists to focus on the final on-screen look of a character, rather than working indirectly on the abstract pose shapes.

3 ANIMATING IN PREMO

Premo's internal graph execution model had to be updated in order to support performant editing and execution of full-resolution rigs. This work was detailed in LibEE2 [Bryson and Papp 2018]. This dual

graph representation enables Premo to load a full-fidelity editable production rig, and translate it on-the-fly into a fast animation representation. This unlocks a number of important workflows. For example, artists are now able to receive interactive feedback during character development - a rig change can immediately be "played back" in-session, with no build step required. Conversely, animation workflows gain increased access to authoring capabilities. An animator can work with a rigger to spot-fix a problem live, while animating, without having to leave their workflow.

4 RESULTS

Premo's new rigging-enabled architecture was first deployed on the animated holiday special *Trolls: Holiday in Harmony*. This gave the artists a chance to mirror five existing characters and see them in the same production context as their production-vetted counterparts. Moving from legacy rigging software to the Premo architecture was a major advancement for production, taking advantage of a modern interface, multithreaded performance, and a robust Python API. By leaning on the data model, riggers were able to push out changes to characters quickly, and install wardrobe elements in a drag-and-drop manner, a process which used to require a large amount of text-based data manipulation. The experience on *Trolls: Holiday in Harmony* was a production testing and adoption runway for expanded use of rigging in Premo on the upcoming feature film *Trolls: Band Together* where it was used to rig 26 characters. This larger adoption allowed for production-scale use of the software, and to continue to refine the user experience and day-to-day rigging workflows. Riggers appreciate the ability to work in the same software used by animators. This enables riggers to analyze bugs in a production shot context and push fixes back to the rig to address the issue in all shots.

Premo's new rigging-enabled architecture is the foundation of our feature character creation platform, enabling artists to build high-performance characters in a sophisticated and forward-looking framework. By delivering this robust experience to our character riggers, previously only available to our animators, we have developed a novel paradigm for character rigging that solidifies decades of feature character experience into a data model, set of workflows, and scene visualization to create the future of characters at DreamWorks.

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